

INFT-1205 – Data Communications & Networking II

Capstone Project

Datacore Router Configuration

Institution: Durham College

By: Group 1

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Document Overview

This document serves as a brief documentation on the Datacore Router. It includes an overview of the Datacore router, along with its configuration.

Device Overview

The DatacoreRouter is the core router for Datacore Solutions. It handles inter-VLAN routing using Router-on-a-Stick, provides DHCP services for IT and HR, and allows WAN connectivity to the Primary Rogers ISP and the Backup Bell ISP using static and floating static routes.

Configuration

Building configuration...

Current configuration : 5205 bytes

!

! Last configuration change at 13:06:56 UTC Wed Apr 8 2026

!

version 17.6

service timestamps debug datetime msec

service timestamps log datetime msec

platform qfp utilization monitor load 80

platform punt-keepalive disable-kernel-core

platform hardware throughput crypto 15M

!

hostname DatacoreRouter

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

!

!

!

!

no ip domain lookup

ip domain name Datacore.com

ip dhcp excluded-address 10.160.0.1 10.160.0.10

ip dhcp excluded-address 10.160.1.1 10.160.1.10

!

ip dhcp pool IT

network 10.160.0.0 255.255.255.128

default-router 10.160.0.1

!

ip dhcp pool HR

network 10.160.1.0 255.255.255.192

default-router 10.160.1.1

!

!

!

login on-success log

!

!

!

!

!

!

!

subscriber templating

!

!

!

!

!

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

!

!

crypto pki trustpoint TP-self-signed-1266970963

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-1266970963

revocation-check none

rsakeypair TP-self-signed-1266970963

!

!

crypto pki certificate chain TP-self-signed-1266970963

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 31323636 39373039 3633301E 170D3236 30343038 31313232
34375A17 0D333630 34303731 31323234 375A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 32363639
37303936 33308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 0100955E DBBD8F69 3CB1D525 58B90313 06646C81 D9F69B37 5B5DD344
91751C32 AACEC93C 46D29FC2 42533637 16537966 4753B160 6CF9E57A A8652359
1B7E377B 35C9C45F 76216669 998495CA 32497753 A2B025A3 FC8BC5FB F3A890DF
DD941AC2 753018C3 1AAA4D53 990437A2 702685DA 6797E206 ED4A5D11 41DB5330
8750D126 19C895B5 D3E289D8 D792DCFF A141D1F5 512C9C70 1F57AC32 642BD3E6
8ABE8AFB 57C4887E E3BE43B8 C7E9C136 EC8B67C2 9CF59F9A AF849602 57E9C45C
79CA5DC4 AB7AD069 1C2F95AD 033DB7B0 E46D9691 20B804F9 5349FBEC 7FEB7808
18DD0EA1 CCED3F4D BDDEB3DA 74C2BD03 12984781 14714251 AC78E00A 2AEB3A79
405FD848 B01D0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 1464C08B C6B89B93 1B16E5A3 8AAEE127 AF970AE4
28301D06 03551D0E 04160414 64C08BC6 B89B931B 16E5A38A AEE127AF 970AE428
300D0609 2A864886 F70D0101 05050003 82010100 75E4786E 014BFA53 C99E5C45
6BAE8D30 822AF5F5 64E6DC8A 3F4000C7 52A0BEF4 EC32BFBB 726084C5 A6302507
5E81FB0B 67B93FA0 E6EB33CB 112348FB 318E64E2 CD8871E5 78715315 F6B70A9D
8288C649 FDC87563 D93553DE 0DD616F4 600181F8 883ADD1E E17415FF A19057C3
179504F0 6AA08E45 168341C5 5954E762 D99B9F29 F304E2DA 21CBD558 559F33B0
75D66B39 550D0D5F A589B08B 78DFD6DD 1ED9F676 CC8C8DBC 259A1753 CD127962
783C9029 6F20D36E 422D1E59 4069104D 87E8365E 54A86A80 8A1A291A 6DFB498F
9CF902F8 FEC3DE2B 991D4F01 D81C1D76 C65DEC0D A8E05EBB 0E6C4E77 D5F7A323

0DC1E1F8 76E859DB 95E1E77B 7434899D C1917DDF

quit

!

!

!

!

!

!

!

!

!

license udi pid C8200L-1N-4T sn FJC27291G9P

license boot level network-advantage addon dna-advantage

memory free low-watermark processor 67482

!

diagnostic bootup level minimal

!

spanning-tree extend system-id

!

username DatacoreAdmin secret 9

\$9\$rWUIz/JwHu4re.\$fjR3tkl1mJHlMRdEWdeSfb.5PdqhhdW1ruYMb7ma6nHQ

!

redundancy

mode none

!

!

! !

```
interface GigabitEthernet0/0/0
```

no ip address

negotiation auto

!

interface GigabitEthernet0/0/0.10

description Sub-interface for IT VLAN

encapsulation dot1Q 10

ip address 10.160.0.1 255.255.255.128

!

interface GigabitEthernet0/0/0.20

description Sub-interface for Customer Service VLAN

encapsulation dot1Q 20

ip address 10.160.0.129 255.255.255.192

!

interface GigabitEthernet0/0/0.30

description Sub-interface for Sales and Marketing VLAN

encapsulation dot1Q 30

ip address 10.160.0.193 255.255.255.192

!

interface GigabitEthernet0/0/0.40

description Sub-interface for the HR VLAN

encapsulation dot1Q 40

ip address 10.160.1.1 255.255.255.192

!

interface GigabitEthernet0/0/0.50

description Sub-interface for the Management VLAN

encapsulation dot1Q 50

ip address 10.160.1.193 255.255.255.224

!

```
interface GigabitEthernet0/0/0.60
description Sub-interface for System Admins 60
encapsulation dot1Q 60
ip address 10.160.1.65 255.255.255.192
```

!

```
interface GigabitEthernet0/0/0.70
description Sub-interface for Accounting 70
encapsulation dot1Q 70
ip address 10.160.1.129 255.255.255.192
```

!

```
interface GigabitEthernet0/0/0.80
description Sub-interface for Legal 80
encapsulation dot1Q 80
ip address 10.160.1.225 255.255.255.224
```

!

```
interface GigabitEthernet0/0/0.1000
description Native VLAN
encapsulation dot1Q 1000 native
```

!

```
interface GigabitEthernet0/0/1
no ip address
negotiation auto
```

!

```
interface GigabitEthernet0/0/2
no ip address
```

negotiation auto

!

interface GigabitEthernet0/0/3

no ip address

negotiation auto

!

interface Serial0/1/0

description primary link to Rogers ISP1

ip address 101.1.1.1 255.255.255.252

clock rate 256000

!

interface Serial0/1/1

description back up link to bell ISP2

ip address 101.1.2.1 255.255.255.252

!

ip http server

ip http authentication local

ip http secure-server

ip forward-protocol nd

ip route 0.0.0.0 0.0.0.0 101.1.1.2

ip route 0.0.0.0 0.0.0.0 101.1.2.2 10

ip route 101.1.1.0 255.255.255.252 Serial0/1/0

ip route 101.1.2.0 255.255.255.252 Serial0/1/1 10

ip ssh version 2

!

!

!

!

!

!

!

control-plane

!

!

mgcp behavior rsip-range tgcp-only

mgcp behavior comedia-role none

mgcp behavior comedia-check-media-src disable

mgcp behavior comedia-sdp-force disable

!

mgcp profile default

!

!

!

!

!

banner motd ^CC Unauthorized Access is Prohibited ^C

!

line con 0

password 7 054F211D2E595E583A2A39

login

stopbits 1

line aux 0

```
line vty 0 4
login local
transport input ssh
line vty 5 14
login
transport input ssh
!
!
!
!
!
!
end
```